

Name: Mariya Krutnyk

20

Any attempt to use an electronic device with internet access (cellphone, tablet, smartwatch, etc.) during this exam will be regarded as cheating. UIC considers cheating a serious violation of academic integrity, with penalties typically ranging from a failing grade on the exam or in the course to suspension or expulsion.

TRANSPORT PHENOMENA II, CHE 312

EXAM 2

CLOSED BOOK, TIME: 75 MINUTES

Solid species A in the shape of a sphere of radius R is submerged in a large body of stagnant liquid B. Species A is slightly soluble in the liquid B with an equilibrium concentration of c_{A0} at $r = R$. The dissolved species A diffuses radially in the liquid phase while it undergoes a first-order chemical reaction.

Answer the following questions in terms of the following constants: radius of the solid sphere R , the diffusivity D_{AB} , and the rate constant k_1''' .

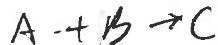
1. Using a shell mole balance for species A, derive an ordinary differential equation for the steady-state concentration profile of species A in the liquid phase. Assume $x_A \ll 1$

2. Solve the above differential equation and express the concentration profile of species A in terms of the

Thiele modulus $\phi \equiv \sqrt{\frac{k_1'''}{D_{AB}}} R$ and two integration constants. Hint: $\frac{1}{r} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) = \frac{d^2}{dr^2} (rc_A)$

3. Determine the integration constants using two appropriate boundary conditions for the concentration profile.

4. Determine how many moles of A react per unit time.



$$\int_S \frac{d}{dt} \rho dS = - \int_S \langle N_A \rangle dS + \int_V R_A dV$$

1. $0 = - \int_S \langle N_A \rangle dS$

can't cancel
 $0 = [4\pi r^2 N_A]_r - [4\pi r^2 N_A]_{r+dr}$
 $0 = \frac{dN_A}{dr}$

$-p D_{AB} \frac{dc_A}{dr} = c_A v$

$\frac{d}{dr} \left(-p D_{AB} \frac{dc_A}{dr} = c_A v \right)$

$-p D_{AB} r \frac{d^2 c_A}{dr^2} = \frac{dc_A}{dr} r v + C_1$

$0 = -p D_{AB} \frac{dc_A}{dr} - c_A v$

$0 = -p D_{AB} r \frac{d^2 c_A}{dr^2}$

$0 = -p D_{AB} \frac{d^2}{dr^2} (rc_A)$

$N_A = -p D_{AB} \frac{dc_A}{dr}$

$N_A = c_A v$

$R_A = c_A k_1'''$

$0 = \frac{dc_A}{dr} - s.s.$

$c_A(R) = c_{A0}$

$\Gamma = \frac{1}{A} \left(\frac{d}{dr} \cos \phi + \sin \phi \right)$

Page 1 of 3
 $\Gamma = \frac{1}{A} \left(\frac{d}{dr} \cos \phi + \sin \phi \right)$

$\frac{V}{r}$
 $\frac{2}{L} \frac{L}{2}$



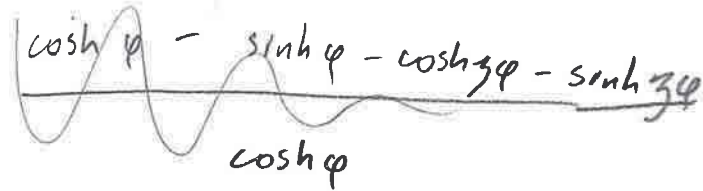
r, θ, ϕ

$V = \frac{4}{3} \pi r^3$

Name: Marissa Krumholz

Any attempt to use an electronic device with internet access (cellphone, tablet, smartwatch, etc.) during this exam will be regarded as cheating. UIC considers cheating a serious violation of academic integrity, with penalties typically ranging from a failing grade on the exam or in the course to suspension or expulsion.

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\Gamma = \cosh \varphi - \sinh \varphi - \cosh 3\varphi - \sinh 3\varphi$$


$$0 = X(N_{Ar} + N_{Br}) + CV^4$$

$$0 = -CD_{AB} \frac{dC_A}{dr} - k_1''' X_A$$

$$CD_{AB} \frac{dC_A}{dr} = -k_1''' X_A$$

$$\frac{1}{r} \frac{d}{dr} \left(r CD_{AB} \frac{dC_A}{dr} \right) = k_1''' C X_A$$

$$CD_{AB} \frac{d^2}{dr^2} (rC_A) = k_1''' C \frac{dX_A}{dr}$$

$$\frac{D_{AB}}{k_1'''} \frac{d^2}{dr^2} (rC_A) = \frac{dX_A}{dr}$$

$$\frac{D_{AB}}{k_1'''} \frac{d^2}{dr^2} (rC_A) = 0$$

$$\frac{D_{AB}}{k_1'''} \frac{d}{dr} (rC_A) = C_1 \quad C_A = C_{A0} \sqrt{\frac{k_1'''}{D_{AB}}} \left(\cosh \frac{R}{r} - \sinh \frac{R}{r} \right)$$

$$\frac{D_{AB}}{k_1'''} \frac{d}{dr} (rC_A) = C_1 r + C_2$$

Name:

Any attempt to use an electronic device with internet access (cellphone, tablet, smartwatch, etc.) during this exam will be regarded as cheating. UIC considers cheating a serious violation of academic integrity, with penalties typically ranging from a failing grade on the exam or in the course to suspension or expulsion.

$$\langle n_A \rangle = C_A V + j_A$$

$$j_A = C_{AB} \frac{dx_A}{dr}$$
$$j_A = \rho_{AB} \frac{dc_A}{dr}$$